Question-Level Survey Consolidation Analysis

Federal Survey Concept Mapper Project

Brock Webb

2026-01-28

Table of contents

Preface

This report documents a proof-of-concept analysis examining whether federal survey questions can be consolidated through record linkage with the American Community Survey (ACS).

Key Finding

~11% of conceptually overlapping questions are actually consolidable.

The gap between concept-level overlap and question-level consolidation reflects real methodological differences:

- Reference period mismatches
- Construct differences (same topic, different operationalization)
- Screener vs. battery design patterns

Scope

- **Surveys analyzed**: FoodAPS, CPS (compared against ACS)
- **Question pairs classified**: 1,702
- **Method**: Dual-LLM classification with inter-rater validation

Data

Source data and analysis scripts are available in the project repository.

Chapter 1

Overview

1.1 Project Context

This report documents the methodology and findings from a question-level consolidation analysis comparing federal surveys with the American Community Survey (ACS). The goal is to determine realistic consolidation potential through record linkage by examining actual question pairs rather than relying solely on concept-level overlap.

1.2 Analysis Process

The analysis follows a four-phase approach:

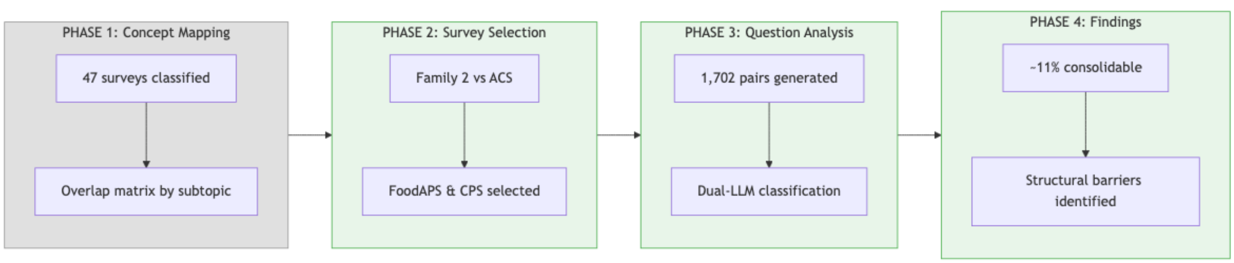

Figure 1.1: Process Flow

Phase 1: Concept Mapping (Prior Work) - Classified 47 federal surveys using Census Bureau topic taxonomy - Generated overlap matrices by subtopic - Identified survey "families" with shared conceptual domains

Phase 2: Survey Selection (This Analysis Begins) - Focused on "Family 2" - Economic Household Surveys - Compared 5 surveys against ACS: SIPP, CE, AHS, CPS, FoodAPS - Selected FoodAPS (123 overlap) and CPS (181 overlap) as pilot cases due to manageable question counts

Phase 3: Question Analysis - Generated 1,702 question pairs within shared subtopics - Applied dual-model LLM classification (Claude Haiku 4.5 + GPT-5-mini) - Classified pairs as consolidable, partially consolidable, or not consolidable

Phase 4: Findings - ~11% overall consolidation potential (structural ceiling) - Identified three primary barriers: construct mismatch, reference period incompatibility, screener vs. battery design - Validated methodology through case studies (SNAP, Disability, Employment Status)

1.3 Key Finding

Concept overlap is a ceiling, not an estimate.

The 123 FoodAPS questions sharing subtopics with ACS yielded only 74 consolidable pairs (12.1%). The 181 CPS questions yielded 118 consolidable pairs (10.8%). The ~90% gap between concept overlap and actual consolidation potential reflects real methodological differences in how surveys operationalize the same constructs.

1.4 Report Structure

Section	Content
1. Survey Selection	Why Family 2, why FoodAPS & CPS
2. Concept Overlap	Treemap visualization of subtopic distribution
3. Methodology	Classification workflow, prompts, decision logic
4. Results	Consolidation rates, agreement metrics, data tables
5. Case Studies	Deep-dives: SNAP, Race, Disability, Employment
6. Synthesis	Cross-survey patterns, barrier taxonomy, recommendations
7. Future Work	Next surveys, extensions, limitations

Chapter 2

Survey Selection: Family 2 Economic Household Surveys

2.1 The ACS as a Linkage Backbone

The American Community Survey (ACS) is the largest household survey in the federal statistical system, collecting data continuously from approximately 3.5 million households annually. Its comprehensive coverage of demographics, economics, housing, and social characteristics makes it a natural candidate for record linkage with specialized surveys.

The key insight: rather than each survey independently collecting the same demographic information, surveys could potentially sample from ACS respondents and link records, collecting only the specialized content unique to their mission.

2.2 Family 2: Economic Household Surveys

From the broader concept mapping analysis (Phase 1), we identified "Family 2" as surveys sharing substantial conceptual overlap with ACS in economic and household domains:

Survey	Full Name	Total Shared Questions	Subtopics Covered	Dominant Domain
SIPP	Survey of Income and Program Participation	577	38	Economic (304)
AHS	American Housing Survey	460	34	Housing (343)
CE	Consumer Expenditure Survey	283	30	Housing/Economic
CPS	Current Population Survey	181	25	Economic (110)
FoodAPS	Food Acquisition and Purchase Survey	123	23	Economic (59)

Total across Family 2: 1,624 shared questions with ACS

2.3 Why FoodAPS and CPS?

We selected FoodAPS and CPS as pilot cases for question-level analysis:

Practical considerations: - Lowest question counts (123 and 181) → tractable for methodology development - Combined: 1,702 question pairs to classify - Estimated API cost: ~$1.50 total

Analytical considerations: - Different overlap profiles: FoodAPS is SNAP-heavy, CPS is employment-heavy - Different survey designs: FoodAPS is a specialized supplement, CPS is a core labor force survey - Testing generalizability across survey types

What we deferred: - SIPP (577 questions) - largest, save for scaled analysis - AHS (460 questions) - housing-dominated, different domain - CE (283 questions) - moderate size, future candidate

2.4 Domain Distribution

2.4.1 FoodAPS (123 questions)

Domain	Count	Top Subtopics
Economic	59	Food Stamps/SNAP (22), Earnings (10), Employment (9)
Social	42	Household (18), School Enrollment (11), Veterans (5)
Demographic	12	Age (3), Sex (3), Race (2), Relationship (2)
Housing	10	Costs (6), Vehicles (2), Tenure (1)

2.4.2 CPS (181 questions)

Domain	Count	Top Subtopics
Economic	110	Employment Status (36), Earnings (23), Hours/Week (19)
Social	41	Disability (17), Household (10), Education (5)
Demographic	29	Relationship (10), Age (3), Race (3), Hispanic (3)
Housing	1	Tenure (1)

2.5 Implications for Pair Generation

Questions are paired **within shared subtopics only**. This means: - FoodAPS's 22 SNAP questions pair with ACS's 1 SNAP question → 22 pairs - CPS's 36 Employment Status questions pair with ACS's 6 Employment questions → 216 pairs - Cross-subtopic comparisons are excluded (reduces noise)

The total pair counts: - **FoodAPS × ACS**: 610 pairs - **CPS × ACS**: 1,092 pairs - **Combined**: 1,702 pairs

This many-to-many pairing within subtopics is why pair counts exceed question counts.

Chapter 3

Concept-Level Overlap Visualization

3.1 Understanding the Treemaps

Before examining individual question pairs, we visualize where each survey's overlap with ACS concentrates. These treemaps show:

- **Box size**: Number of questions in that subtopic sharing conceptual overlap with ACS
- **Box color**: Domain category (Economic, Social, Housing, Demographic)
- **Labels**: Subtopic name and question count

3.2 FoodAPS-ACS Concept Overlap

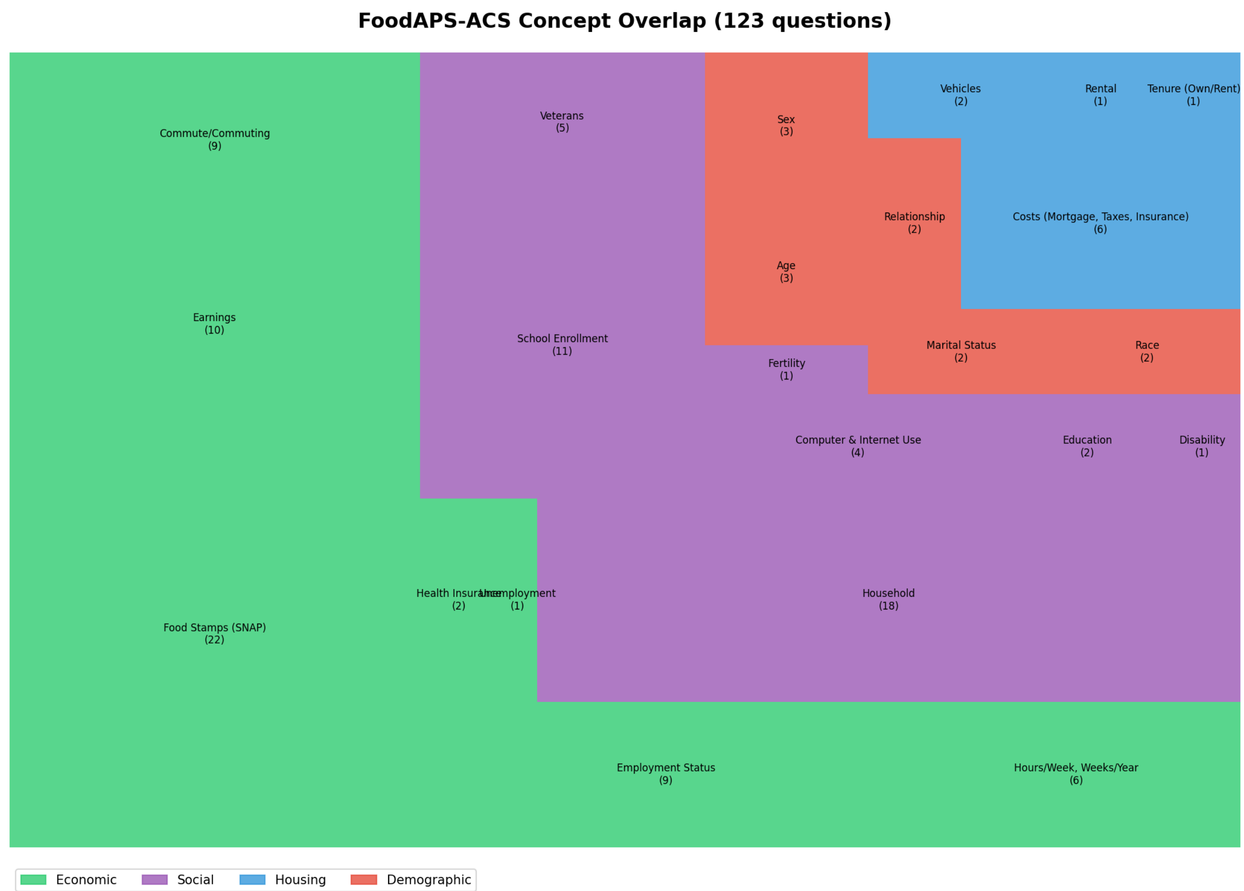

Figure 3.1: FoodAPS-ACS Treemap

Key observations:

1. **SNAP dominates Economic domain** (22 of 59 economic questions)
   - FoodAPS was designed specifically for food assistance research
   - This concentration creates many pairs but most are screener-vs-battery mismatches
2. **Household structure is significant** (18 questions in Social)
   - Understanding who lives together matters for food acquisition patterns
   - These tend to consolidate well with ACS
3. **Demographics are sparse but important** (12 questions)
   - Basic demographics (age, sex, race) for each household member
   - High consolidation potential - stable characteristics
4. **Housing is peripheral** (10 questions)
   - Mostly costs and vehicle access (relevant for food access)
   - Not FoodAPS's primary focus

3.3 CPS-ACS Concept Overlap

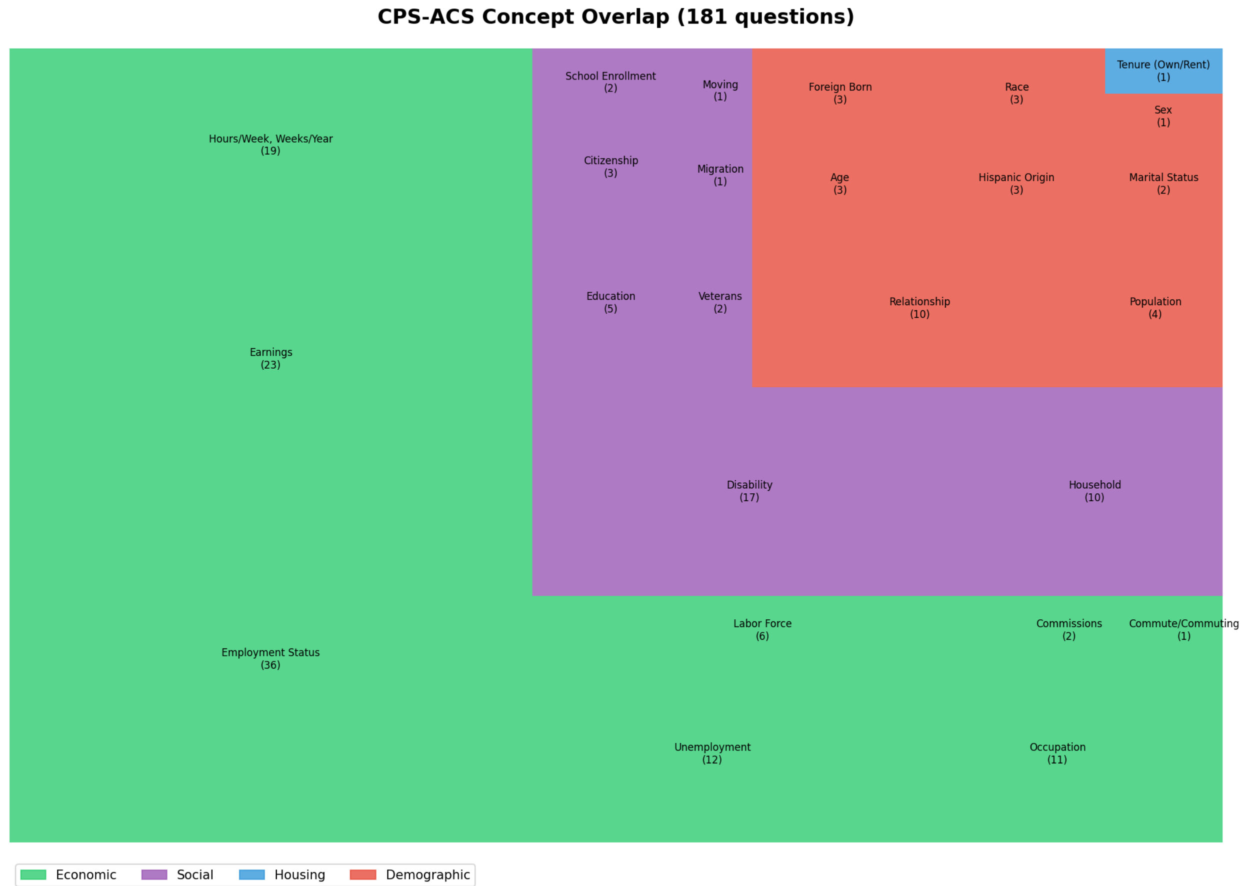

Figure 3.2: CPS-ACS Treemap

Key observations:

1. **Employment dominates** (36 questions)
   - Core CPS mission: monthly labor force statistics
   - Many reference period mismatches with ACS's "last week" framing
2. **Disability is substantial** (17 questions in Social)
   - CPS includes both work-limiting disability (Type A) and ACS6 functional questions (Type B)
   - Construct mismatch is significant - same topic, different operationalization
3. **Demographics are richer than FoodAPS** (29 questions)
   - More detailed relationship, nativity, and migration questions
   - Better consolidation potential
4. **Housing is minimal** (1 question - tenure)
   - CPS is not a housing survey
   - Almost no housing overlap with ACS

3.4 What These Treemaps Tell Us

Concept overlap is necessary but not sufficient for consolidation.

The treemaps show WHERE overlap exists, but not WHETHER questions can substitute for each other. Consider:

Subtopic	Concept Overlap	Consolidation Reality
SNAP (FoodAPS)	22 questions	Only 2 consolidable (8.7%) - different reference periods, depths
Employment (CPS)	36 questions	~4 consolidable (11%) - reference period mismatches
Race (FoodAPS)	2 questions	2 consolidable (100%) - stable demographic
Disability (CPS)	17 questions	6 consolidable (35%) - only ACS6-compatible subset

The gap between treemap area and consolidation potential IS the finding.

This gap reflects: 1. Reference period incompatibility 2. Construct differences (same concept, different operationalization) 3. Screener vs. battery design (breadth vs. depth)

The question-level analysis quantifies this gap precisely.

3.5 Interpreting Domain Colors

Domain	Color	Typical Consolidation	Why
Economic	Green	10-15%	Reference periods rarely align; surveys need different temporal granularity
Social	Purple	20-40%	Mixed - household composition consolidates, disability depends on construct
Housing	Blue	5-15%	Specialized needs; surveys focus on different housing aspects
Demographic	Red	60-100%	Stable characteristics; standardized across surveys

The demographic domain consistently shows highest consolidation potential because: - Characteristics don't change between survey administrations - Constructs are standardized (OMB definitions for race/ethnicity) - No reference period complications (age, sex are point-in-time facts)

Economic and employment questions face the steepest barriers because: - Labor force status requires specific temporal windows - Different surveys need different windows for their analytical purposes - "Last week" "last 4 weeks" "last 12 months" - these serve different uses

Chapter 4

Classification Methodology

4.1 Workflow Overview

This methodology uses dual-LLM classification to assess whether question pairs from different surveys measure the same construct with compatible reference periods and response formats.

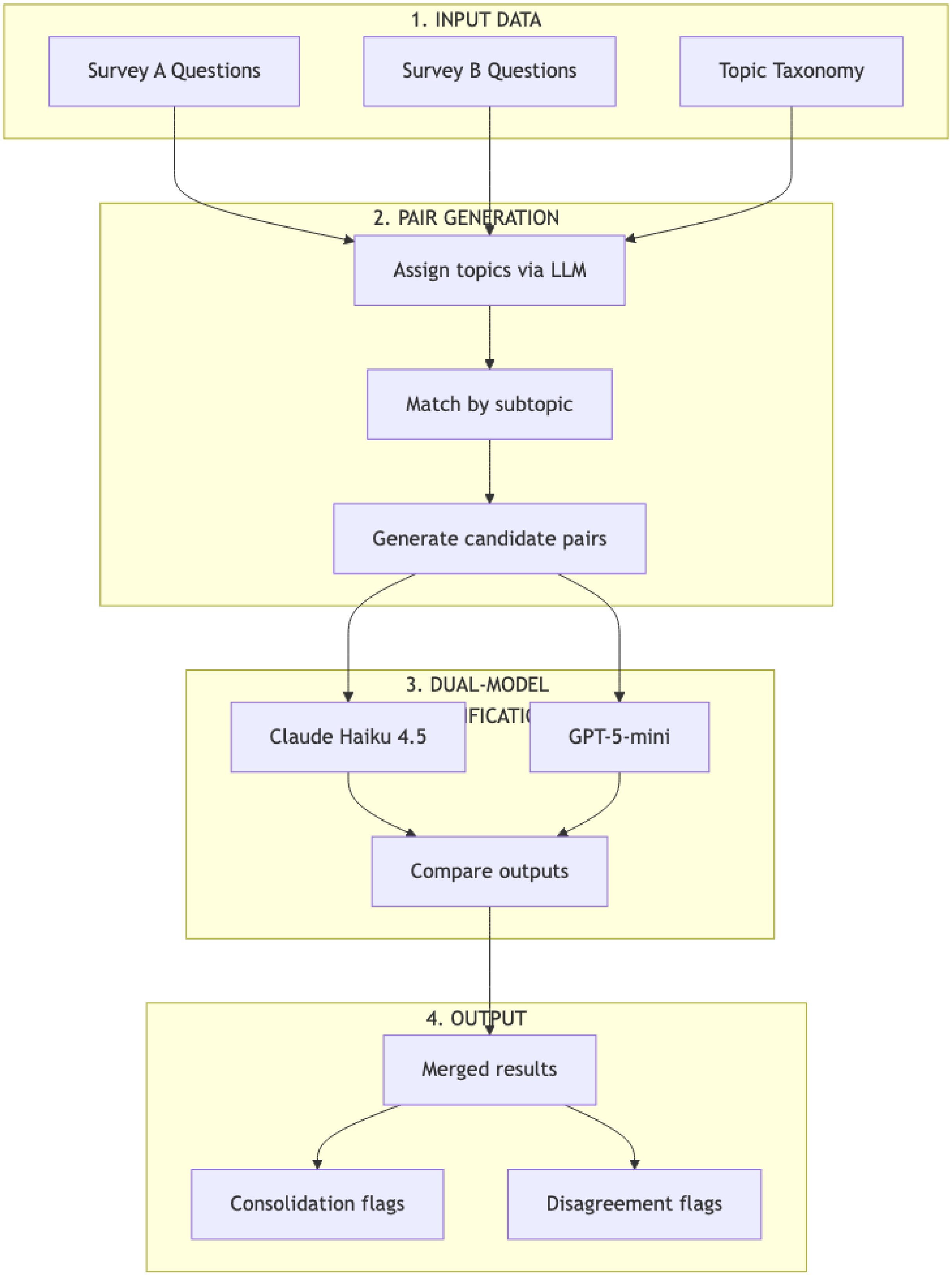

Figure 4.1: Classification Workflow

4.2 Stage 1: Input Data

4.2.1 Source Questions

Questions are extracted from official survey instruments:

Survey	Source	Questions Extracted
FoodAPS	USDA FoodAPS questionnaire PDFs	610 unique questions
CPS	Census Bureau CPS instrument	~400 unique questions
ACS	Census Bureau ACS instrument	~150 unique questions

4.2.2 Topic Taxonomy

We use the Census Bureau's official topic taxonomy with standardized categories:

Top-level topics: Demographics, Economics, Housing, Social

Subtopics (examples): - Demographics → Age, Sex, Race, Hispanic Origin, Citizenship - Economics → Employment Status, Hours/Week, Earnings, Unemployment - Social → Disability, Veterans, School Enrollment

4.3 Stage 2: Pair Generation

4.3.1 Topic Assignment

Each question is classified into the taxonomy using LLM:

```
INPUT:  "How many hours do you usually work per week at your main job?"
OUTPUT: Topic: Economics
        Subtopic: Hours/Week, Weeks/Year
        Confidence: 0.95
```

4.3.2 Pair Matching Logic

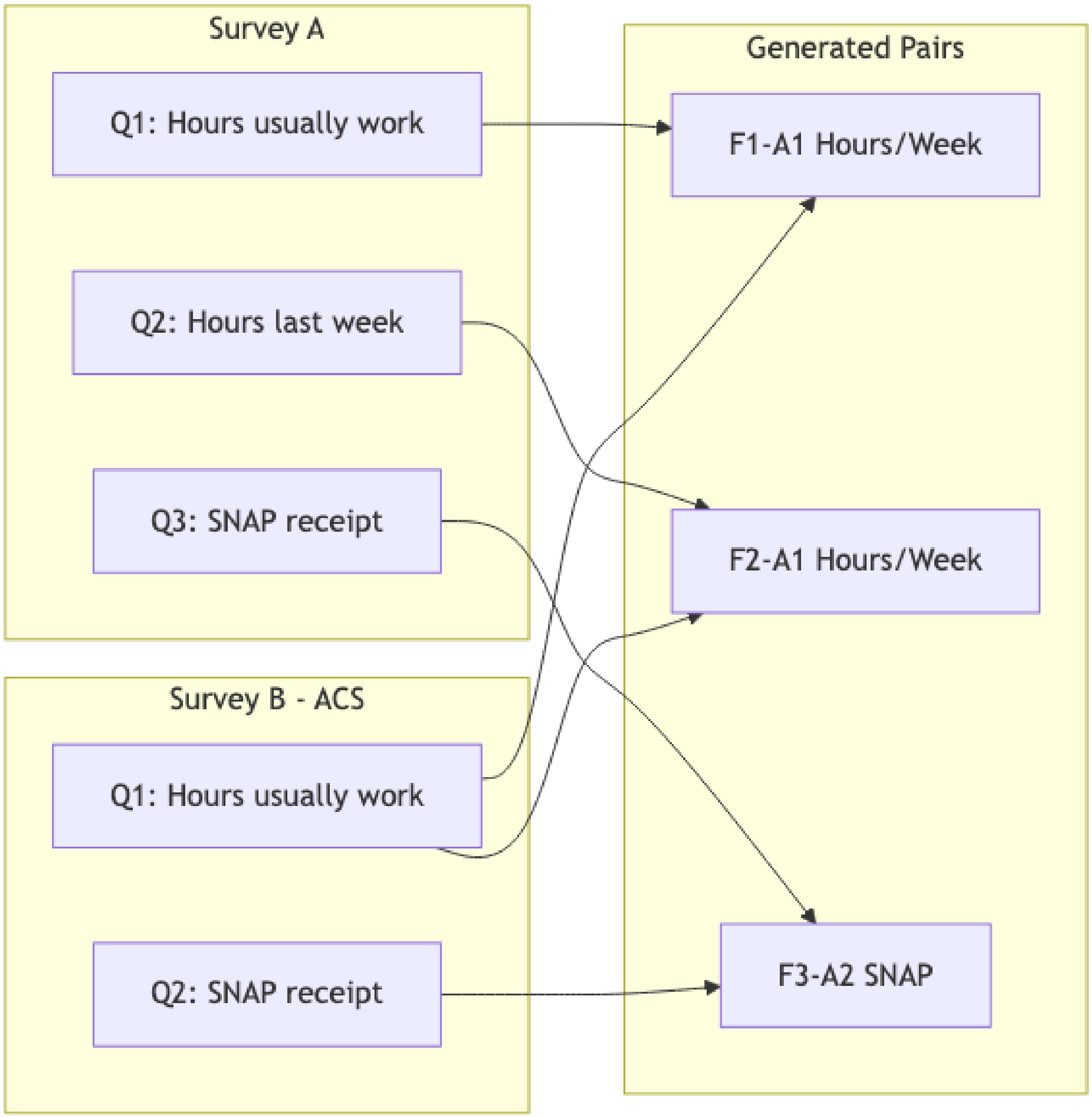

Figure 4.2: Pair Matching

Questions are paired **within shared subtopics only**: - Only questions with **matching subtopics** are paired - This is a **many-to-many** relationship within subtopics - Cross-subtopic pairs are not generated (reduces noise)

Example pair counts:

Subtopic	FoodAPS Qs	ACS Qs	Pairs Generated
Hours/Week	8	7	56 (8×7)

Subtopic	FoodAPS Qs	ACS Qs	Pairs Generated
SNAP	23	1	23 (23 × 1)
Disability	2	6	12 (2 × 6)
Sex	27	1	27 (27 × 1)

4.4 Stage 3: Dual-Model Classification

4.4.1 Classification Schema

Each model assigns one of six classifications:

Classification	Definition	Consolidation Potential
`exact_duplicate`	Identical wording, format, reference period	**Yes**
`near_duplicate`	Minor wording differences, same construct	**Yes**
`related_but_distinct`	Same topic, different specifics	No
`reference_period_mismatch`	Same construct, incompatible time windows	No
`response_format_mismatch`	Same construct, incompatible response types	No
`not_comparable`	Different constructs or topics	No

4.4.2 Classification Prompt

Each question pair is sent to both models with identical prompts:

```
You are analyzing two survey questions to determine if they measure
the same construct and could potentially be consolidated through
record linkage.

QUESTION A (from {Survey A}):
"{question_a_text}"

QUESTION B (from {Survey B}):
"{question_b_text}"

Analyze these questions and provide:

1. CLASSIFICATION: Choose exactly one from the schema above
2. CONFIDENCE: high, medium, or low
3. REASONING: 2-3 sentences explaining your classification
4. CONSOLIDATION_POTENTIAL: yes, partial, or no
5. REFERENCE_PERIOD_A: Extract the time frame from Question A
6. REFERENCE_PERIOD_B: Extract the time frame from Question B

Respond in JSON format.
```

4.4.3 Model Configuration

Parameter	Claude Haiku 4.5	GPT-5-mini
Temperature	0.0	0.0
Max tokens	500	500
Batch size	10 questions	10 questions
Parallel workers	6	6
Rate limiting	Exponential backoff	Exponential backoff

Why two models? - Reduces single-model bias - Disagreements flag ambiguous cases - Agreement increases confidence - Different training data → different blind spots

4.5 Stage 4: Output and Analysis

4.5.1 Consolidation Determination

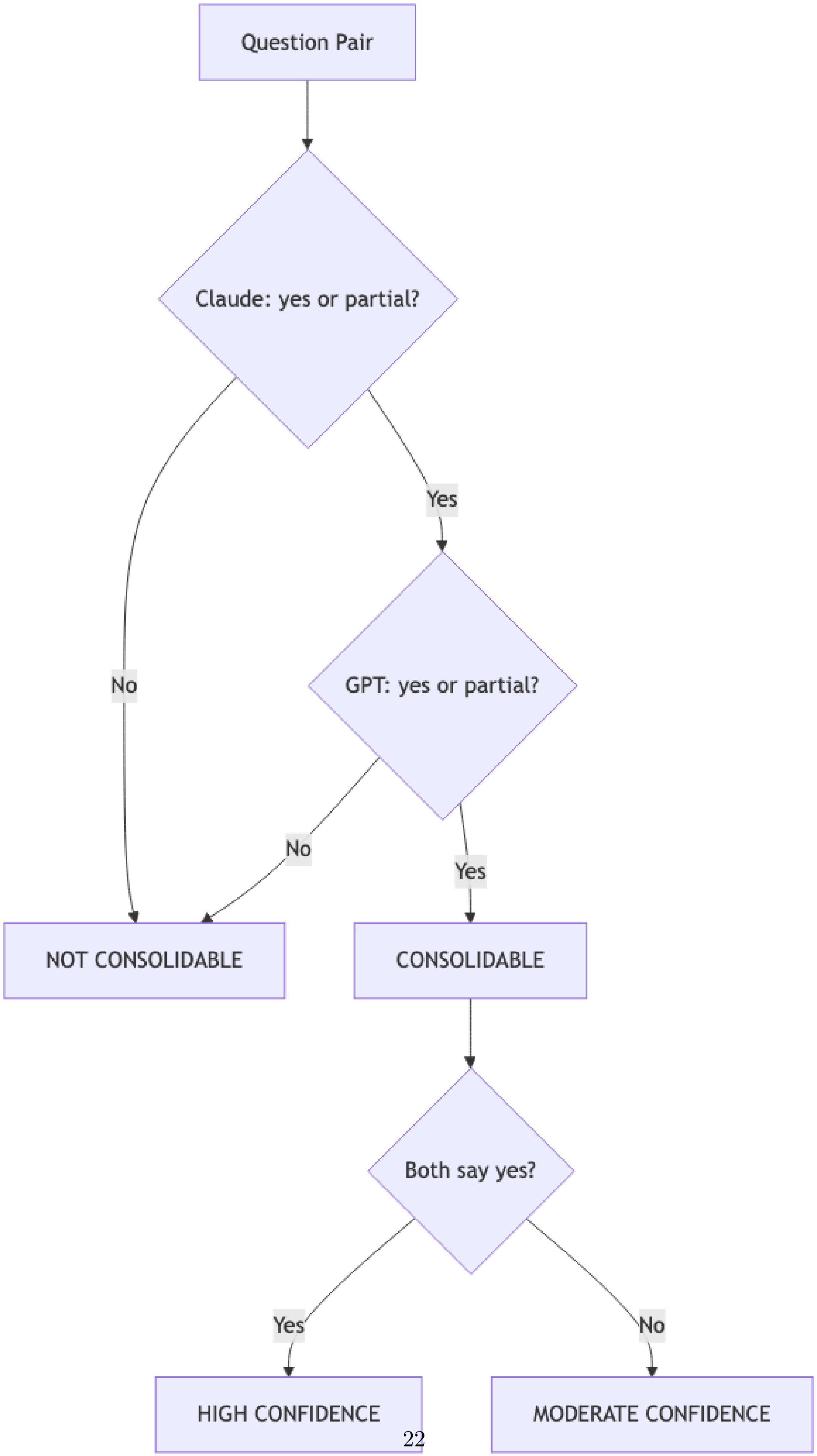

Figure 4.3: Decision Tree

A pair is flagged as **consolidable** when:

Decision rules: - Both models say YES → High confidence, likely droppable **- Both models say PARTIAL** → Moderate confidence, partial overlap **- One YES, one PARTIAL** → Moderate confidence, needs review **- Any NO** → Not consolidable (conservative approach)

4.5.2 Disagreement Handling

Disagreement Type	Action
Classification differs, consolidation same	Use consolidation consensus
Consolidation differs	Flag for human review
One high confidence, one low	Weight toward high confidence

4.6 Classification Criteria

4.6.1 Exact Duplicate Criteria

All must be true: - Same core construct (what is being measured) - Same or equivalent wording - Same response format (binary, numeric, categorical) - Same reference period (or both unspecified) - Same unit of analysis (person, household)

4.6.2 Near Duplicate Criteria

Most must be true: - Same core construct - Similar but not identical wording - Compatible response format - Same or compatible reference period - Same unit of analysis

4.6.3 Reference Period Mismatch Criteria

- Same core construct
- Compatible wording
- **Different reference periods that serve different purposes**

4.6.4 Construct Mismatch (Not Comparable) Criteria

- **Different constructs despite same topic domain**

4.7 Quality Assurance

4.7.1 Automated Checks

Check	Purpose	Action if Failed
JSON parse	Valid model output	Retry with backoff
Required fields	Complete response	Retry or flag
Enum validation	Valid classification	Reject and retry
Confidence present	Quality signal	Default to "medium"

4.7.2 Checkpoint/Resume

Long runs use checkpoint files. If interrupted, processing resumes from last checkpoint.

4.8 Interpreting Results

4.8.1 Consolidation Rate Calculation

```
Consolidation Rate = (Pairs where BOTH models say yes/partial) / Total Pairs
```

4.8.2 What "Consolidable" Means

It means: If you had perfect person-linkage between surveys, this question could potentially be dropped from Survey A because Survey B already collects equivalent data.

It does NOT mean: - The questions are identical - Linkage is feasible - Data quality would be equivalent - Skip logic permits removal

4.8.3 Model Agreement Interpretation

Agreement Rate	Interpretation
>90%	Clear cases, high confidence in results
70-90%	Normal range, some genuine ambiguity
<70%	Many borderline cases, interpret cautiously

4.9 Reproducibility

4.9.1 Parameters

```
MODELS = ["claude-haiku-4.5", "gpt-5-mini"]
TEMPERATURE = 0.0
MAX_TOKENS = 500
BATCH_SIZE = 10
PARALLEL_WORKERS = 6
REQUIRE_BOTH_MODELS = True
CONSOLIDATION_VALUES = ["yes", "partial"]
```

4.9.2 Cost Tracking

Survey Pair	Pairs	Total Cost
FoodAPS-ACS	610	~$0.50
CPS-ACS	1,092	~$1.00
Total	**1,702**	**~$1.50**

4.10 Limitations

4.10.1 What LLMs Can't Assess

1. **Skip logic dependencies** - Whether removing a question breaks survey flow
2. **Analytical continuity** - Impact on time series if source changes
3. **Response quality** - Whether linked data has same error properties
4. **Linkage feasibility** - Whether person-matching is actually possible

4.10.2 When Human Review Is Required

- Model disagreement on consolidation potential
- High-stakes consolidation decisions
- Unusual or domain-specific constructs
- Questions with complex skip logic

Chapter 5

Results

5.1 Executive Summary

5.1.1 Key Finding: ~11% Consolidation Potential, Structurally Bounded

Survey Pair	Total Pairs	Both Models Agree Consolidable	Rate
FoodAPS-ACS	610	74	12.1%
CPS-ACS	1,092	118	10.8%
Combined	**1,702**	**192**	**11.3%**

Both surveys converge on ~11% regardless of their different missions (food behavior vs labor force). This convergence suggests a **structural ceiling** rather than survey-specific limitations.

5.1.2 Why Only 11%?

1. **Demographics consolidate well** (60-100%): Sex, age, race are stable characteristics with standardized constructs
2. **Habitual measures partially consolidate** (~26%): "Usually/normally" framing aligns across surveys

3. **Point-in-time measures rarely consolidate** (~4-12%): Reference period mismatches block substitution
4. **Construct mismatches block consolidation entirely** (0-2%): Same topic, different operationalization

5.1.3 LLM Validation: High Accuracy on Clear Cases

The CPS Disability analysis provides strong validation: - **6/6 ACS6 standardized question matches identified** (perfect diagonal) - **336/336 non-matching pairs correctly rejected** - Model agreement: 94.7% on Disability subtopic

---

5.2 Detailed Results

5.2.1 FoodAPS-ACS (610 pairs)

5.2.1.1 Classification Distribution

Classification	Claude Haiku 4.5	GPT-5-mini
related_but_distinct	283 (46%)	419 (69%)
not_comparable	179 (29%)	53 (9%)
near_duplicate	44 (7%)	28 (5%)
response_format_mismatch	38 (6%)	32 (5%)
reference_period_mismatch	36 (6%)	39 (6%)
exact_duplicate	30 (5%)	39 (6%)

5.2.1.2 Consolidation by Subtopic

Subtopic	Consolidable/Total	Rate	Pattern
Education	2/2	100%	Demographic
Tenure (Own/Rent)	1/1	100%	Demographic
Sex	24/27	89%	Demographic
Race	**5/6**	**83%**	Demographic
Age	6/9	67%	Demographic
Relationship	3/6	50%	Demographic
Hours/Week, Weeks/Year	16/56	29%	Habitual framing
Marital Status	2/8	25%	Demographic
Veterans	5/36	14%	Mixed
Employment Status	6/60	10%	Point-in-time
SNAP	**2/23**	**8.7%**	Screener vs battery
Earnings	3/90	3%	Reference period
School Enrollment	2/65	3%	Reference period
Costs	1/54	2%	Construct mismatch
Household	1/72	1%	Construct mismatch
Disability	0/12	0%	Construct mismatch
Commute	0/55	0%	Reference period
Computer/Internet	0/12	0%	Construct mismatch
Health Insurance	0/6	0%	Construct mismatch

5.2.2 CPS-ACS (1,092 pairs)

5.2.2.1 Classification Distribution

Classification	Claude Haiku 4.5	GPT-5-mini
related_but_distinct	693 (63%)	781 (72%)
not_comparable	141 (13%)	19 (2%)
near_duplicate	83 (8%)	54 (5%)
reference_period_mismatch	77 (7%)	71 (7%)
response_format_mismatch	63 (6%)	103 (9%)
exact_duplicate	35 (3%)	64 (6%)

5.2.2.2 Consolidation by Subtopic

Subtopic	Consolidable/Total	Rate	Pattern
Sex	3/3	100%	Demographic

Subtopic	Consolidable/Total	Rate	Pattern
Hispanic Origin	6/9	67%	Demographic
Occupation	10/17	59%	Standardized codes
Commissions	1/2	50%	Small N
Education	2/5	40%	Demographic
Age	3/9	33%	Demographic
Citizenship	1/3	33%	Demographic
Relationship	8/30	27%	Demographic
Hours/Week, Weeks/Year	44/168	26%	Habitual framing
Veterans	2/8	25%	Demographic
Race	2/9	22%	Demographic
Foreign Born	1/6	17%	Demographic
Earnings	17/138	12%	Reference period
Marital Status	1/8	12%	Demographic
Population	1/8	12%	Response format
School Enrollment	1/10	10%	Reference period
Employment Status	**25/215**	**11.6%**	Reference period
Unemployment	1/26	4%	Reference period
Disability	**6/342**	**1.8%**	Construct mismatch
Household	0/52	0%	Construct mismatch
Labor Force	0/16	0%	Reference period
Commute	0/5	0%	Reference period

5.3 Structural Barriers to Consolidation

5.3.1 1. Construct Mismatch

Definition: Questions address the same topic but operationalize different constructs serving different analytical purposes.

Topic	Survey A Construct	Survey B Construct	Substitutable?
Disability	Work-limiting	Functional limitation	No
SNAP	Program mechanics	Program prevalence	No
Employment	Labor force flows	Point-in-time status	Partial

Implication: Topic-level overlap analysis dramatically overestimates consolidation potential.

5.3.2 2. Reference Period Incompatibility

Reference Type	Consolidation Potential
Habitual (“usually/normally”)	High - no temporal anchor
Same specific window	High - direct comparison
Overlapping windows	Partial - subset relationship
Non-overlapping windows	Low - different statistics

5.3.3 3. Screener vs. Battery Design

Surveys collect the “same thing” at different depths: - **Screener:** “Did anyone receive SNAP?” (yes/no) - **Battery:** “How much? When? Which cards? Who’s covered?”

ACS optimizes for breadth (many topics, shallow). Specialized surveys optimize for depth (few topics, detailed). These are complementary designs.

5.3.4 4. Response Format Differences

Even identical constructs may use incompatible formats: - Count vs binary ("How many?" vs "Any?") - Roster vs aggregate ("Who specifically?" vs "Anyone?") - Continuous vs categorical ("Exact amount?" vs "Range?")

5.4 Methodological Findings

5.4.1 LLM Classification Performance

Metric	FoodAPS-ACS	CPS-ACS
Model agreement	68.2%	75.4%
Agreement on consolidable	74/74 (100%)	118/118 (100%)
Disability diagonal accuracy	N/A	6/6 (100%)

Interpretation: When both models agree on consolidation, confidence is high. Disagreements indicate genuine ambiguity warranting human review.

5.4.2 Model Generation Effects

Newer models (Claude Haiku 4.5, GPT-5-mini) showed **lower consolidation rates** than earlier models tested during development. This pattern is consistent with improved nuance handling: - Better detection of reference period mismatches - Better discrimination of construct differences - More conservative on borderline cases

5.4.3 Sampling vs. Full Population

Approach	CPS Consolidation Rate	Model Agreement
300-pair stratified sample	15%	66%
Full 1,092-pair run	10.8%	75.4%

Stratified sampling overestimated consolidation by 4 percentage points due to demographic over-representation. Full runs cost ~$0.50-1.00 per survey - sampling offers no meaningful savings.

5.5 Consolidation Patterns by Content Type

Content Type	Typical Consolidation Rate	Rationale
Core demographics (sex, age, race)	60-100%	Stable characteristics, standardized constructs
Habitual measures (hours/week)	25-30%	"Usually/normally" framing aligns

Content Type	Typical Consolidation Rate	Rationale
Point-in-time status	10-15%	Reference period mismatches
Program-specific content	0-10%	Specialized needs require specialized questions

---

5.6 Data Sources

File	Contents	Pairs
`data/foodaps_comparison_merged.csv`	FoodAPS-ACS classifications	610
`data/cps_comparison_merged.csv`	CPS-ACS classifications	1,092

5.6.1 Classification Schema

- **exact_duplicate:** Identical wording, format, reference period
- **near_duplicate:** Minor wording differences, same construct
- **related_but_distinct:** Same topic, different specifics
- **reference_period_mismatch:** Same construct, incompatible time windows
- **response_format_mismatch:** Same construct, incompatible response types
- **not_comparable:** Different topics or constructs

5.6.2 Consolidation Definition

A pair is "consolidable" if **both models** classify it as exact_duplicate, near_duplicate, or assign consolidation_potential = "yes" or "partial".

Part I

Case Studies

Chapter 6

Case Study: FoodAPS-ACS

6.1 Overview

This case study provides detailed analysis of specific FoodAPS-ACS question pairs to illustrate the structural barriers to survey consolidation. Through examination of actual question text and LLM reasoning, we demonstrate that low consolidation rates in specialized content areas are not analytical failures but **expected outcomes of survey design differences**.

Key findings: - **SNAP (8.7% consolidable)**: ACS screener vs. FoodAPS program battery - fundamentally different analytical purposes - **Race (83-100% consolidable)**: Demographics work well - stable characteristics with no temporal scope issues - **Reference periods drive consolidation**: Habitual framing ("usually/normally") enables consolidation; point-in-time framing blocks it

---

6.2 SNAP (Food Stamps) - 2/23 Consolidable (8.7%)

6.2.1 The Paradox

FoodAPS is explicitly designed to study food acquisition behavior among SNAP participants. Yet only 2 of 23 SNAP-related question pairs show consolidation potential with ACS. This appears anomalous - shouldn't the survey's core mission area show *more* overlap with ACS, not less?

6.2.2 The Data

Metric	Value
Total SNAP pairs	23
Claude consolidable	6 (26%)
GPT consolidable	5 (22%)
Both agree consolidable	**2 (8.7%)**
Model agreement rate	48%

6.2.3 What Consolidates: Identical Constructs

Pair FOODAPS_0594 (Both models: YES)

```
FoodAPS: "Did (you/anyone at this address) receive benefits from SNAP in the last 12 months?"
ACS:     "IN THE PAST 12 MONTHS, did you or any member of this household receive benefits
          from the Food Stamp Program or SNAP (the Supplemental Nutrition Assistance Program)?"
```

Why it works: - Identical reference period (12 months) - Same unit of analysis (household-level) - Same construct (binary SNAP receipt)

6.2.4 What Doesn't Consolidate: Screener vs. Battery

Problem Type 1: Reference Period Mismatch

```
FoodAPS: "Have you received benefits from SNAP in the past 30 days?"
ACS:     "IN THE PAST 12 MONTHS, did you receive SNAP benefits?"
```

30 days 12 months. FoodAPS needs **current participation status** to interpret food acquisition diaries.

Problem Type 2: Response Format Mismatch

```
FoodAPS: "How many EBT cards are issued to people at this address?"
ACS:     "Did you receive SNAP benefits?" (yes/no)
```

Count vs binary - ACS provides no information about enrollment intensity.

Problem Type 3: Administrative Detail vs. Prevalence

```
FoodAPS: "Select the names of the people that receive SNAP benefits on this card."
ACS:     "Did you receive SNAP benefits?" (yes/no)
```

Roster identification vs prevalence - entirely different analytical purposes.

6.2.5 The Structural Pattern

FoodAPS Question Type	Count	ACS Substitutable?
12-month receipt screener	1	Yes
Last receipt date	1	Partial
30-day receipt	2	No
Number of EBT cards	1	No
Card assignment roster	~15	No
Benefit amount/timing	~3	No

Conclusion: The 8.7% consolidation rate is structurally determined. FoodAPS needs program *mechanics*; ACS measures program *reach*. These are complementary, not redundant.

6.3 Race/Ethnicity - 5/6 Consolidable (83%)

6.3.1 Why Race Works

The pattern:

```
FoodAPS: "What is your race and/or ethnicity? Select all that apply."
ACS:     "What is Person 1's race?"
```

Why consolidation works: 1. **No temporal scope** - Race is a stable characteristic 2. **Same construct** - Both measure racial/ethnic self-identification 3. **Similar response format** - Both allow multiple selections

Why "partial" instead of "yes": - FoodAPS combines race AND ethnicity in one question - ACS asks race and Hispanic origin separately (standard federal practice)

6.3.2 Demographics: The Reliable Consolidation Target

Demographic Item	FoodAPS Rate	CPS Rate
Sex	89% (24/27)	100% (3/3)
Age	67% (6/9)	33% (3/9)
Race	83% (5/6)	22% (2/9)
Relationship	50% (3/6)	27% (8/30)

6.4 Hours/Week - 16/56 Consolidable (29%)

6.4.1 Why This Works Better Than Employment Status

The key is **habitual framing**.

Consolidable pairs use habitual language:

```
FoodAPS: "How many hours do you NORMALLY work for pay?"
ACS:     "How many hours do you USUALLY work per week?"
```

No specific reference period → direct comparison possible.

Non-consolidable pairs use point-in-time language:

```
FoodAPS: "How many hours did you work LAST WEEK?"
ACS:     "During the PAST 12 MONTHS, did this person usually work 35+ hours?"
```

Weekly snapshot   annual typical hours.

6.4.2 Reference Period Alignment Drives Consolidation

Framing Type	Consolidation Rate
Habitual ("usually/normally")	~40-50%
Point-in-time ("last week")	~10%
Annual retrospective ("past 12 months")	~15%

6.5 Summary: What FoodAPS-ACS Teaches Us

1. **Specialized Surveys Have Specialized Needs** - FoodAPS can't use ACS SNAP data because ACS doesn't collect program mechanics
2. **Demographics Are the Reliable Target** - Race, sex, age work because they're time-invariant and federally standardized
3. **Habitual Framing Enables Cross-Survey Comparison** - "Usually work" is comparable; "worked last week" is not
4. **Low Consolidation Rates Are Features, Not Bugs** - The 8.7% SNAP rate reflects correct survey design

Chapter 7

Case Study: CPS-ACS

7.1 Overview

This case study analyzes CPS-ACS question pairs, focusing on two subtopics that reveal fundamental barriers to survey consolidation:

1. **Disability (1.8% consolidable)**: A textbook case of construct mismatch - CPS measures work-limiting disability while ACS measures functional limitations. The 6 consolidable pairs are exactly the ACS6 standardized questions, validating LLM accuracy.
2. **Employment Status (11.6% consolidable)**: Reference period incompatibility - CPS uses multiple temporal windows for labor force dynamics while ACS uses fixed "last week" snapshots.

7.2 Disability - 6/342 Consolidable (1.8%)

7.2.1 The Structure

The Disability subtopic contains 342 pairs formed from: - **57 unique CPS disability questions** × **6 unique ACS disability questions**

CPS has two fundamentally different types of disability questions.

7.2.2 CPS Disability Question Types

Type A: Work-Limiting Disability (51 questions)

```
"Does your disability prevent you from accepting any kind of work
 during the next 6 months?"
```

Purpose: Labor force participation measurement for BLS employment statistics

Type B: Functional Limitations - ACS6 Compatible (6 questions)

```
CPS_294: "Are you deaf or do you have serious difficulty hearing?"
CPS_295: "Are you blind or do you have serious difficulty seeing?"
CPS_296: "Do you have serious difficulty concentrating, remembering, or making decisions?"
CPS_297: "Do you have serious difficulty walking or climbing stairs?"
CPS_298: "Do you have difficulty dressing or bathing?"
CPS_299: "Do you have difficulty doing errands alone?"
```

Purpose: ADA compliance, accessibility planning (matches ACS purpose)

7.2.3 LLM Validation: Perfect Diagonal Match

The 6 consolidable pairs form a **perfect 1:1 diagonal** between CPS Type B and ACS questions:

CPS Question	ACS Question	Construct	Result
CPS_294	ACS_84	Hearing	CONSOLIDABLE
CPS_295	ACS_85	Vision	CONSOLIDABLE
CPS_296	ACS_86	Cognition	CONSOLIDABLE
CPS_297	ACS_87	Walking	CONSOLIDABLE
CPS_298	ACS_88	Dressing	CONSOLIDABLE
CPS_299	ACS_89	Errands	CONSOLIDABLE

Accuracy: - 6/6 true positive diagonal matches (100%) - 336/336 true negatives - off-diagonal and Type A correctly rejected - Model agreement: 94.7%

This is the strongest validation of LLM classification accuracy in the dataset.

7.2.4 Why Work-Limiting Functional Limitations

Dimension	Work-Limiting	Functional
Question	"Can you work?"	"Can you walk?"
Use case	BLS unemployment stats	ADA compliance
Reference	Forward (next 6 months)	Current/ongoing
Derivability	Not related	Not related

7.3 Employment Status - 25/215 Consolidable (11.6%)

7.3.1 The Reference Period Problem

Metric	Value
Total pairs	215
Both agree consolidable	25 (11.6%)
Reference period mismatches	37 (17.2%)
Model disagreements	**60 (27.9%)**

The 28% disagreement rate is the **highest of any subtopic**.

7.3.2 Reference Period Distribution

CPS uses multiple windows: - Last week / week before last - Last 4 weeks - Last month - Last 12 months - Currently/ongoing

ACS uses one window: - Last week (74%)

The mismatch is structural: CPS needs multiple temporal windows for labor force dynamics.

7.3.3 What Consolidates: Overlapping Windows

```
CPS: "LAST WEEK, did you do ANY work for pay?"
ACS: "LAST WEEK, did this person work for pay at a job?"
```

Same reference period → consolidable.

7.3.4 What Doesn't Consolidate: Incompatible Windows

```
CPS: "Did you do any work for pay in the LAST 4 WEEKS?"
ACS: "LAST WEEK, did this person work for pay?"
```

Different time periods, different statistics → not consolidable.

7.3.5 Model Disagreement Analysis

Disagreement Type	Count	Pattern
Claude YES, GPT NO	13	Claude more lenient on overlapping windows
Claude NO, GPT YES	27	GPT more lenient on construct similarity
Same consolidation, different reasoning	20	Agree on bottom line

Example of legitimate disagreement:

```
CPS: "Do you currently want a job, either full or part time?"
ACS: "LAST WEEK, was this person on layoff from a job?"
```

Job-seeking intention vs layoff status - both models defensible.

7.4 Consolidation Rate by Content Type

Content Type	CPS Rate	Driver
Demographics (sex, age, race)	40-100%	Stable, standardized
Habitual measures (hours/week)	26%	"Usually" framing aligns
Point-in-time status (employment)	11.6%	Reference period mismatch
Construct-specific (disability)	1.8%	Different definitions

7.5 Summary: What CPS-ACS Teaches Us

1. **LLMs Correctly Identify Standardized Matches** - Perfect ACS6 diagonal validates methodology
2. **Multi-Purpose Surveys Resist Consolidation** - CPS's multi-window design serves BLS needs incompatible with ACS's single window
3. **Construct Mismatch Is Invisible at Topic Level** - "Disability" appears in both surveys but measures different things
4. **High Model Disagreement = Genuine Ambiguity** - 28% disagreement in Employment reflects real uncertainty, not model failure
5. **Consolidation Is Structurally Bounded** - 10.8% is an accurate measurement of design-constrained overlap

Chapter 8

Synthesis and Conclusions

8.1 The Central Finding

After analyzing 1,702 question pairs across two major federal surveys, we find that **survey consolidation through ACS record linkage has real but structurally limited potential**. The convergence of both survey pairs on ~11% consolidation—despite serving vastly different purposes (food acquisition behavior vs. labor force statistics)—suggests this is a **fundamental ceiling** imposed by how federal surveys are designed.

This ceiling is not a failure. It reflects the reality that surveys optimize for different analytical needs, and those differences are encoded in question design.

8.2 What We Learned

8.2.1 1. Topic Overlap Question Substitutability

Level of Analysis	Apparent Overlap	Actual Substitutability
Domain (“both measure employment”)	~80-90%	—
Topic (“both ask about work hours”)	~40-60%	—
Question (actual text comparison)	—	**~11%**

Concept-level analysis dramatically overestimates consolidation potential.

8.2.2 2. Three Structural Barriers Explain Most Non-Consolidation

Barrier	Description	Example
Construct mismatch	Same topic, different operationalization	CPS work-limiting disability vs ACS functional limitations
Reference period incompatibility	Same construct, different time windows	CPS “last 4 weeks” vs ACS “last week”
Screener vs battery	Same topic, different depth	ACS “received SNAP?” vs FoodAPS “how many cards?”

These barriers are **features, not bugs**.

8.2.3 3. Content Type Predicts Consolidation Rate

Content Type	Consolidation Rate	Why
Core demographics	60-100%	Stable characteristics, standardized
Habitual measures	25-30%	"Usually/normally" has no temporal anchor
Point-in-time status	10-15%	Reference periods rarely align
Program-specific	0-10%	Specialized needs require specialized questions

8.2.4 4. LLMs Can Reliably Identify Consolidation Opportunities

The Disability case study provides validation:

Test	Result
True positive: ACS6 diagonal	6/6 (100%)
True negative: Off-diagonal	336/336 (100%)

When constructs genuinely align, LLMs find them.

8.3 Policy Implications

8.3.1 The Optimistic View

11% is not nothing. For a 100-question survey: - ~11 questions could potentially be dropped with ACS linkage - Demographics (5-10 questions typically) are nearly fully consolidable

8.3.2 The Realistic View

11% is a ceiling, not a floor. Actual consolidation will be lower because: - Linkage isn't free (consent, matching error, privacy) - Skip logic creates dependencies - Analytical continuity matters - High-burden questions rarely consolidate

8.3.3 The Strategic View

Target high-value opportunities:

Opportunity	Value	Feasibility
Demographic battery consolidation	High	High
Habitual framing coordination	Medium	Medium
Full survey consolidation	Low	Low

8.4 Recommendations

8.4.1 For Statistical Agencies

1. **Target demographics first** - Near-certain wins with minimal risk
2. **Coordinate framing conventions** - Habitual framing consolidates; point-in-time doesn't
3. **Don't expect specialized content to consolidate** - SNAP mechanics, disability work-limitations, monthly employment flows require specialized questions by design
4. **Use question-level analysis for planning** - Concept-level studies overestimate by 5-10x

8.4.2 For Survey Design

1. **Consolidation potential is a design choice** - Surveys needing ACS-linkable data should adopt ACS constructs
2. **Standardization has tradeoffs** - ACS6 disability consolidates perfectly but doesn't measure work-limitation
3. **Reference periods are load-bearing** - "Last week" vs "last 4 weeks" vs "past 12 months" aren't interchangeable

---

8.5 Methodological Contributions

8.5.1 Question-Level Analysis Framework

We demonstrate that question-level comparison using LLM classification is: - **Feasible:** 1,702 pairs classified for ~$1.50 - **Scalable:** Full-population runs preferred over sampling - **Validated:** ACS6 diagonal provides ground truth

8.5.2 Proposed Taxonomy Addition: Construct Mismatch

Current categories don't capture the Disability pattern. Proposed addition:

construct_mismatch: Questions address the same topic but operationalize different constructs serving different analytical purposes. Cannot be substituted regardless of linkage quality.

8.5.3 Dual-Model Validation

Using two independent LLMs provides: - Confidence scoring (both agree = high confidence) - Ambiguity detection (disagreement = needs review) - Bias mitigation

---

8.6 Limitations

1. **No human validation** - LLM classifications need expert review
2. **Assumed contemporaneous collection** - Real linkage has temporal gaps
3. **Skip logic not evaluated** - Survey flow dependencies exist
4. **Perfect linkage assumed** - Real matching has error

8.7 Final Thoughts

Survey consolidation has real but limited potential. The ~11% ceiling allows agencies to: - Set realistic expectations - Target high-value opportunities - Avoid costly efforts on incompatible content - Design new surveys with linkage in mind

Survey design encodes analytical purpose at the question level. Two surveys can "measure employment" while asking fundamentally different questions because they serve fundamentally different needs. This is good survey design, not wasteful duplication.

The goal is not to maximize consolidation but to **identify the subset where consolidation genuinely serves both surveys' analytical purposes**—and our analysis shows that subset is real, identifiable, and worth pursuing.

Chapter 9

Future Work

9.1 Immediate Extensions

9.1.1 Additional Survey Pairs

The methodology developed here can be applied to additional Family 2 surveys:

Survey Pair	Estimated Pairs	Expected Pattern
SIPP-ACS	~1,500-2,000	Similar ~10-12% (income/program focus)
CE-ACS	~800-1,200	Lower ~5-8% (expenditure detail)
AHS-ACS	~1,200-1,600	Variable (housing construct differences)

Estimated cost: ~$2-5 total for all three additional surveys

9.1.2 Methodological Improvements

1. **Human validation sample**
    - 100-200 pairs with expert adjudication
    - Calibrate LLM accuracy against ground truth
    - Identify systematic bias patterns
2. **Construct mismatch classifier**
    - Train model to specifically detect this barrier type
    - Distinguish from reference_period_mismatch and related_but_distinct
3. **Cross-survey framing analysis**
    - Systematic comparison of reference period conventions
    - Identify coordination opportunities

9.2 Research Questions

9.2.1 Does Consolidation Potential Vary by Survey Domain?

We analyzed economic household surveys. Other domains may show different patterns:

Domain	Hypothesis
Health surveys (NHIS, MEPS)	Higher - more demographic content
Education surveys (NCES)	Moderate - standardized constructs exist
Business surveys	Lower - specialized definitional needs

9.2.2 Can Reference Period Coordination Increase Consolidation?

The Hours/Week finding (26-29% consolidation due to habitual framing) suggests deliberate coordination could help. Research questions:

- Which surveys could adopt habitual framing without analytical loss?
- What's the cost-benefit of survey redesign vs. accepting current ceilings?

9.2.3 What's the Actual Linkage Error Rate?

Our analysis assumes perfect linkage. Real implementation faces: - Person-matching error - Temporal misalignment (ACS from different month) - Consent refusal - Coverage gaps

A simulation study could model realistic consolidation under imperfect linkage.

9.3 Technical Extensions

9.3.1 Alternative Classification Approaches

- Fine-tuned models on survey question pairs
- Embedding-based similarity (failed for topic classification but might work for direct question comparison)
- Hybrid LLM + rule-based systems

9.3.2 Visualization and Reporting

- Interactive consolidation explorer
- Survey-specific consolidation reports
- Policy briefing documents

9.3.3 Integration with Survey Metadata

- Link to question skip logic
- Incorporate temporal collection windows
- Connect to survey purpose documentation

9.4 Resource Summary

Item	Cost
Completed analysis (FoodAPS + CPS)	~$1.50
Future surveys (SIPP + CE + AHS)	~$3-5
Human validation sample	TBD (analyst time)
Total API costs	<$10

The question-level analysis framework is ready for extension. The primary constraint is analyst time for interpretation, not computational cost.